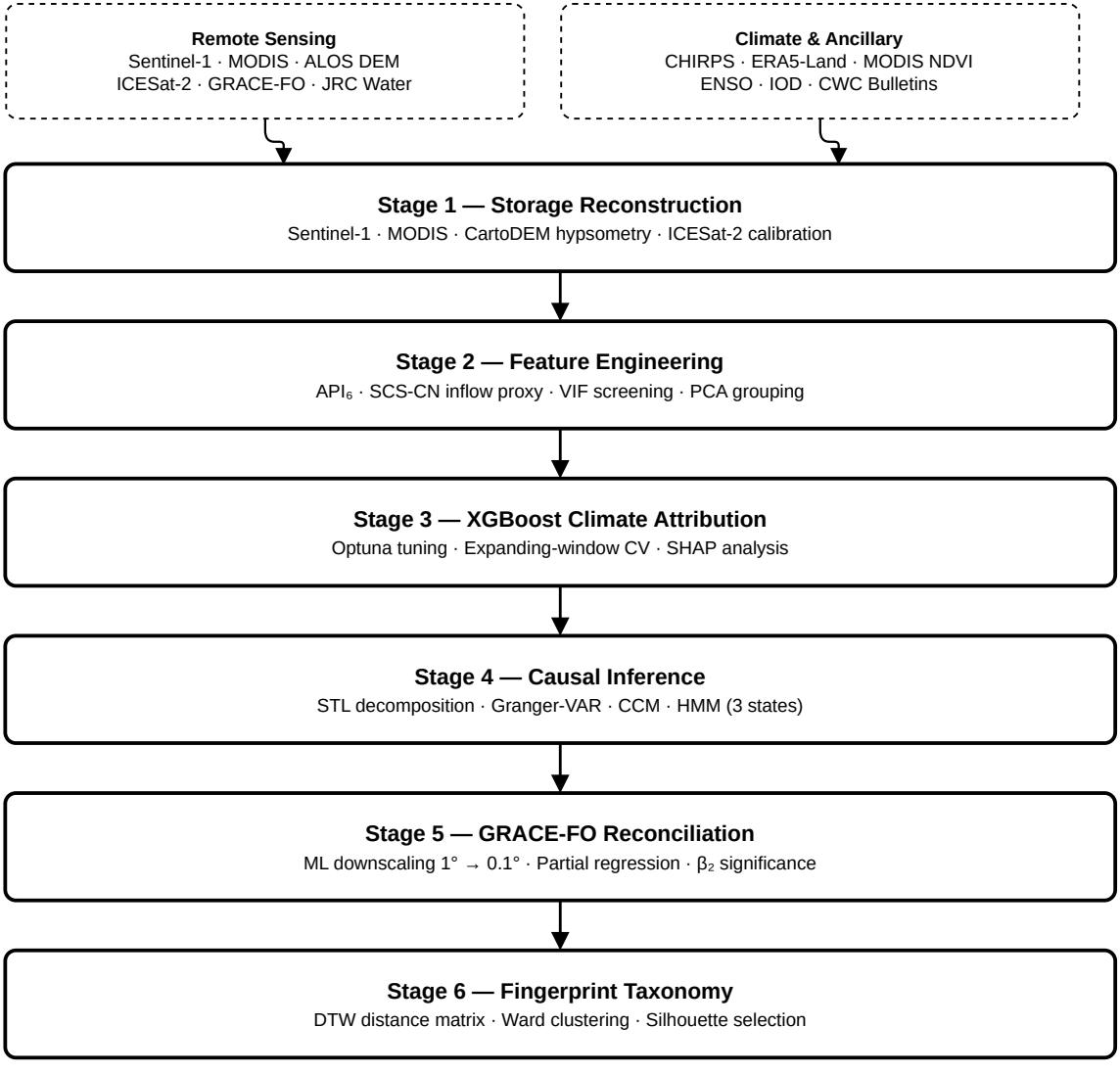
Remote Sensing
Sentinel-1 · MODIS · ALOS DEM
ICESat-2 · GRACE-FO · JRC Water
Climate & Ancillary
CHIRPS · ERA5-Land · MODIS NDVI
ENSO · IOD · CWC Bulletins
Stage 1 — Storage Reconstruction
Sentinel-1 · MODIS · CartoDEM hypsometry · ICESat-2 calibration
Stage 2 — Feature Engineering
API6 · SCS-CN inflow proxy · VIF screening · PCA grouping
Stage 3 — XGBoost Climate Attribution
Optuna tuning · Expanding-window CV · SHAP analysis
Stage 4 — Causal Inference
STL decomposition · Granger-VAR · CCM · HMM (3 states)
Stage 5 — GRACE-FO Reconciliation
ML downscaling 1° → 0.1° · Partial regression · β2 significance
Stage 6 — Fingerprint Taxonomy
DTW distance matrix · Ward clustering · Silhouette selection